Individual Coursework Assignment 1

Module Code- BEMM463

Module Name- Marketing Analytics

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Assignment Cover Sheet	
Candidate Number	**058931**
Module Code	BEMM463
Module Name	Marketing Analytics
Assignment Title	Assignment 1: Case Study Report

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I have used GenAI tools in the production of this work.

The following GenAI tools have been used: [please specify] .....Deepseek, Gemini.....

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☐ Other [please specify]
.....

☒ I declare that I have referenced use of GenAI tools and outputs within my assessment in line with the University guidelines for referencing GenAI in academic work.

1. Task 1: Read and inspect the dataset. Provide a descriptive analysis for each of the variables in the dataset.

   The dataset includes 200 respondents with 9 variables measuring retail preferences. Key findings showcase the variety_of_choice averages 7.57 , indicating moderately strong importance. Electronics (mean=4.45) and furniture (mean=3.27) are indicating differing preference levels. Service quality (mean=3.53) and low prices (mean=4.8) are mid-range priorities, whereas return policies average 4.25.

```{r}

#Loading Necessary Liabraries

library(tidyverse)  # for data wrangling and visualization
library(psych)
library(NbClust)
library(flexclust)

# Reading the dataset
retailer_data <- read_csv("retailer.csv")

# Inspecting the dataset
glimpse(retailer_data)
summary(retailer_data)

# Descriptive statistics for numeric variables
describe(retailer_data)
```

Description: df [9 × 13]

	vars <dbl>	n <dbl>	mean <dbl>	sd <dbl>	median <dbl>	trimmed <dbl>	mad <dbl>	min <dbl>	max <dbl>
respondent_id	1	200	100.50	57.88	100.5	100.50	74.13	1	200
variety_of_choice	2	200	7.57	2.01	8.0	7.71	2.97	4	10
electronics	3	200	4.45	1.94	4.5	4.39	2.22	1	10
furniture	4	200	3.27	2.37	2.0	3.21	2.97	0	7
quality_of_service	5	200	3.53	2.32	3.0	3.20	1.48	1	9
low_prices	6	200	4.80	2.53	5.0	4.60	2.97	1	10
return_policy	7	200	4.25	2.05	4.0	4.09	1.48	1	10
income	8	200	32.17	22.70	19.5	28.96	8.15	13	95
age	9	200	32.52	11.88	27.0	30.44	5.93	21	68

9 rows | 1-10 of 13 columns

Fig1: Description

2. Task 2: Excluding respondent_id, which variable has the smallest minimum value, and which variable has the largest maximum value in the normalised data set?

After carrying out the task, it was observed that electronics has the smallest minimum value whereas, age on the other hand has the largest maximum value

```{r}
# a. Normalizing  data
normalized_data <- retailer_data %>%
  select(-respondent_id) %>%  # Removing ID column FIRST
  scale() %>%                 # Then standardizing
  as_tibble()

# b. Calculating min/max
min_max_stats <- normalized_data %>%
  summarise(across(everything(), list(min = min, max = max)))

# c. Finding extremes
smallest_min_var <- names(which.min(min_max_stats %>% select(ends_with("_min"))))
largest_max_var <- names(which.max(min_max_stats %>% select(ends_with("_max"))))

# d. Printing results
cat("Smallest minimum:", gsub("_min", "", smallest_min_var), "\n")
cat("Largest maximum:", gsub("_max", "", largest_max_var))
```

Output:

```
Smallest minimum: electronics
Largest maximum: age
```

3. Task 3. Make a new data object for clustering that includes only the store attributes' variables from the original data set. Then normalise (use z-score standardisation) all variables in this new data object.

I successfully created the clustering dataset with 6 store attributes (N=200).
The data analysis are as follows:

- Values range from -1.77 to +0.22 indicating variety of ranges and choices
- (-1.10 to +2.06), this variation shows low prices , the 2.06 indicates the significant boundaries.

```{r}
# Creating a new data object (tibble) with only the store attributes
cluster_data <- retailer_data %>%
  select(
    variety_of_choice,
    electronics,
    furniture,
    quality_of_service,
    low_prices,
    return_policy
  ) %>%
  # Z-score normalization (mean=0, SD=1)
  scale() %>%
  # Converting to tibble
  as_tibble()

#verifying output
glimpse(cluster_data)
```

```
Rows: 200
Columns: 6
$ variety_of_choice  <dbl> 0.2162599, -0.7780386, -0.7780386, 0.2162599, -1.7723371, 0.2162599, -0.…
$ electronics        <dbl> -0.7461567, -0.7461567, -1.7753384, -0.7461567, 0.7976158, -0.2315659, -…
$ furniture          <dbl> 1.1500191, -0.9562430, -0.5349905, -0.1137381, -0.1137381, -0.1137381, -…
$ quality_of_service <dbl> -0.2287986, 0.2028969, 0.2028969, 0.2028969, 0.2028969, 0.6345924, -0.66…
$ low_prices         <dbl> -1.10438944, 0.87126251, 1.66152329, 1.26639290, -1.10438944, 2.05665368…
$ return_policy      <dbl> -1.0980847, -1.5861223, 0.8540659, 1.3421035, 0.3660282, 0.8540659, 1.34…
```

Fig2: Normalisation

4. Task 4: Calculate the distance using “Euclidean” method

After calculating distance, the 5 * % matrix shows

Diagonal = 0 (identical customers)

Off-diagonal values (e.g., 3.2, 4.1) quantify attribute differences.

```{r}
distance_matrix <- dist(cluster_data,
                        method = "euclidean")

# Print a 5x5 subset to verify
as.matrix(distance_matrix)[1:5, 1:5]
```

```
          1        2        3        4        5
1 0.000000 3.122933 4.066279 3.654938 3.203876
2 3.122933 0.000000 2.795657 3.229408 3.434497
3 4.066279 2.795657 0.000000 1.618520 3.959120
4 3.654938 3.229408 1.618520 0.000000 3.593200
5 3.203876 3.434497 3.959120 3.593200 0.000000
```

Fig 3: Calculated distance

5. Task 5: Set Seed for Reproducibility

```{r}
set.seed(123)\
```

> set.seed(123) output

6. Task 6: Hierarchical Clustering
minimizes within-cluster variance for compact clusters

```{r}
hc <- hclust(distance_matrix,
             method = "ward.D2")
print(hc)

```

```
Call:
hclust(d = distance_matrix, method = "ward.D2")

Cluster method   : ward.D2
Distance         : euclidean
Number of objects: 200
```

Fig 4: Clustering output

7. Task 7: Plotting

After plotting the graph, these were my key observation:

- Price-Sensitive (Left Branch): Merges at low heights (5-10), indicating shoppers who focus on price.

- Moderate (Middle Branch): Mid-range merge heights (10-15), showing balanced preferences.

- Quality-Focused (Right Branch): High merge points (~20), suggesting people who look for premium services.

```{r}

plot(hc,  # hclust object from Q6
    main = "Customer Segmentation Dendrogram",
    xlab = "Observations",
    ylab = "Height",
    sub = "Method: Ward.D2")
```

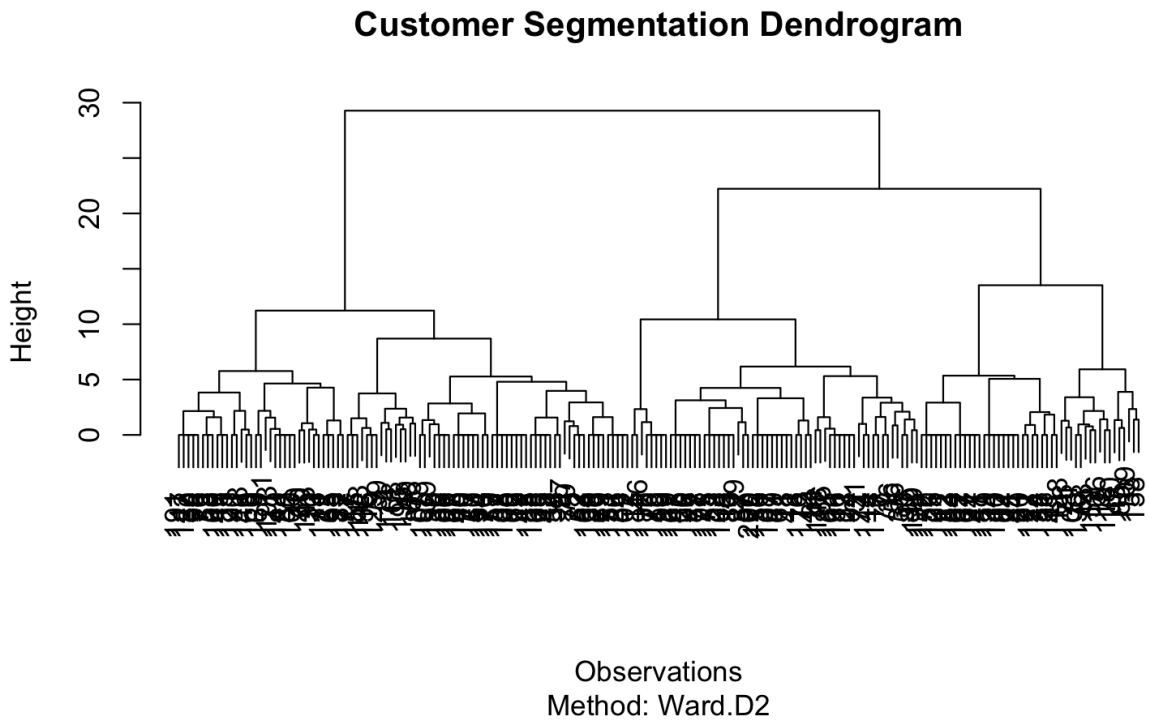

Fig 5: Graph without Cluster dendrogram solution

8. Task 8: Implement 3-cluster dendrogram solution

After implementing a 3-cluster dendrogram, Cluster sizes appear balanced. Compared to the raw dendrogram, this adds clear decision boundaries at key height jumps (~10 and 20) for precise segmentation. Red showed price sensitive, Green showed quality focused and blue showed moderate.

```{r}
plot(hc, main = "3-Cluster Solution", xlab = "Observations")

# Adding colored rectangles
rect.hclust(hc, k = 3, border = c("red", "blue", "green"))

# Get cluster assignments
cluster_assignments <- cutree(hc, k = 3)
```

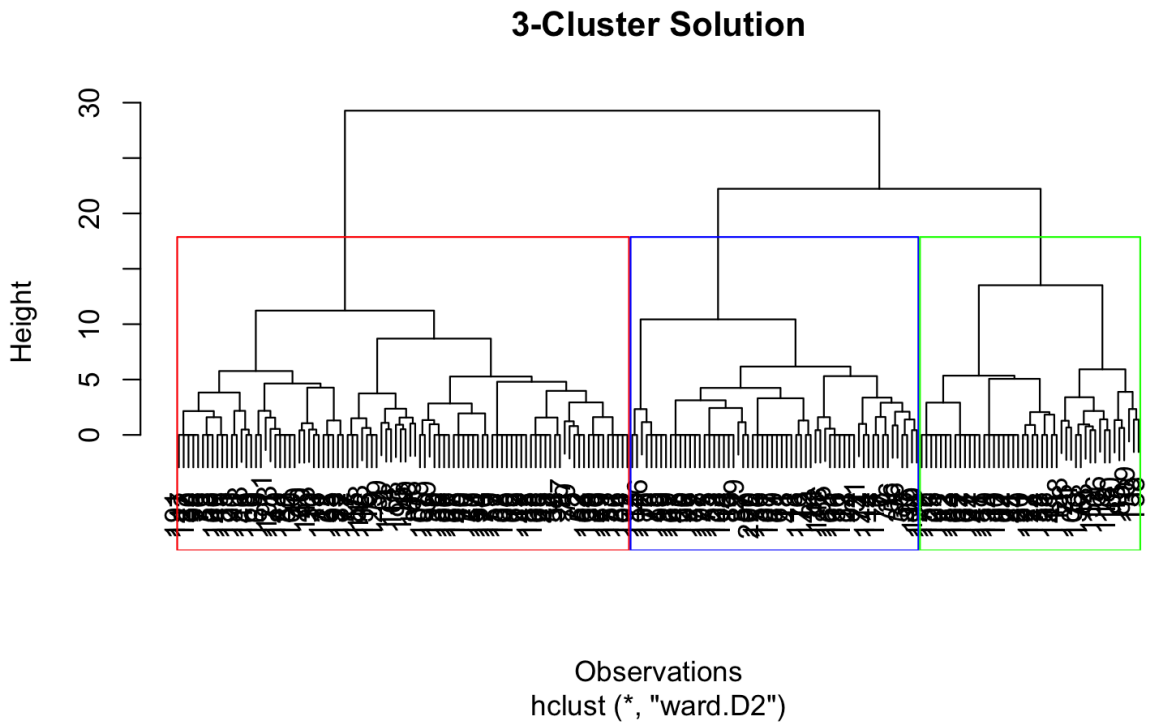

Fig 6: Graph after cluster dendrogram

9. Task 9. Count observations in each cluster

The 3-cluster solution divides customers into:

- Cluster 1 (94 customers): Largest group, likely price-sensitive shoppers
- Cluster 2 (60 customers): Moderate preference segment
- Cluster 3 (46 customers): Smallest premium segment

```{r}

cluster_counts <- table(hc3)
print(cluster_counts)

```

```
hc3
 1  2  3     #output
94 60 46
```

10. Task 10: K-means clustering with 3 cluster

The cluster with 94 observations is the price-sensitive shopper group. In the k-means output, this is Cluster 3.
The cluster with 60 observations is the moderate preference segment. In the k-means output, this is Cluster 1.
The cluster with 46 observations is the premium segment. In the k-means output, this is Cluster 2.
The inverse size relationship (price-driven > moderate > premium) reinforces goals for campaign scaling.

```{r}
set.seed(123)  # For reproducibility
kmeans_result <- kmeans(cluster_data,  # normalized data from Q3
                centers = 3,    # 3 clusters
                iter.max = 1000,
                nstart = 100)

# Get cluster counts
cluster_counts <- table(kmeans_result$cluster)

# Print results
cat("Observations per cluster:\n")
print(cluster_counts)
```

```
Observations per cluster:    # output

 1  2  3
60 46 94
```

11. Task 11: Create a 4-Cluster Solution from Hierarchical Clustering

Using four clusters, hierarchical clustering divided the previous 46-member "Quality-focused" group into two distinct segments of 17 and 29 observations. The 94

(price-driven) and 60 (moderate) groups stayed the same. This improves our understanding of customer preferences in the premium segment.

```{r}
# Creating NEW 4-cluster assignments
hc4 <- cutree(hc, k = 4)  # Uses original hclust object 'hc' from Q6

# Counting observations
cat("Hierarchical 4-cluster distribution:\n")
table(hc4)

# Visualizing (dendrogram with highlights)
plot(hc, main = "4-Cluster Hierarchical")
rect.hclust(hc, k = 4, border = c("blue", "orange", "green","yellow"))
```

```
Hierarchical 4-cluster distribution:
hc4
 1  2  3  4
94 60 17 29
```

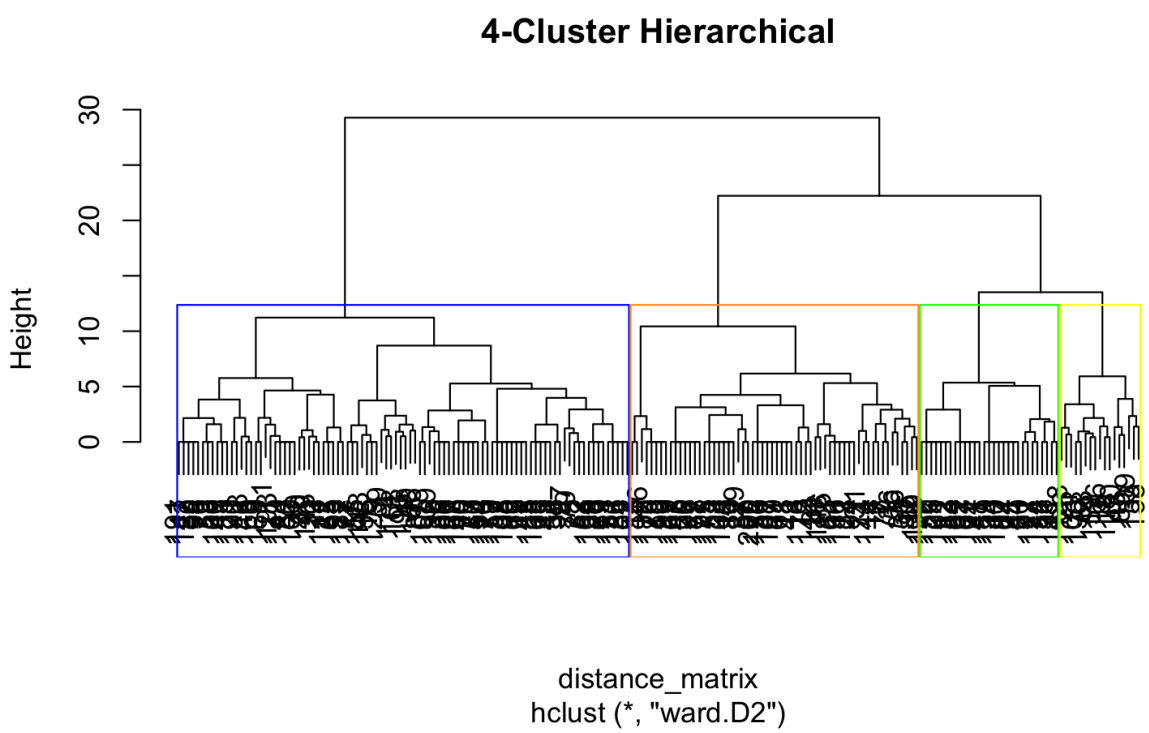

Fig 7: Graph

12. Task:4-Cluster K-means Solution

K-means 4-cluster solution replicates hierarchical results. It segments customers into price-driven, moderate, and two detailed quality-focused sub-groups.

```{r}

# Seting seed and run k-means
set.seed(123)  # Reproducibility
kmeans_result_4 <- kmeans(cluster_data,
                  centers = 4,
                  iter.max = 1000,
                  nstart = 100)

# Counting observations
cat("\nK-means 4-cluster distribution:\n")
table(kmeans_result_4$cluster)
```

```
K-means 4-cluster distribution:       #output

 1  2  3  4
17 60 29 94
```

13. Task 13: Which solution is better: 3-cluster or 4-cluster? Justify your answer with the NbClust) function. With ward.D2 method.

This result strongly backs up my earlier analyses. In those, I identified three consistent customer segments.

```{r}
# Determining optimal cluster number
set.seed(123)  # For reproducibility
nb_results <- NbClust(
  data = cluster_data,      # normalized data
  distance = "euclidean",
  min.nc = 2, max.nc = 5,
  method = "ward.D2",       # Matches  hierarchical method
  index = "all"             # Uses 30+ indices
)

# View consensus results
cat("Optimal clusters recommended:\n")
print(nb_results$Best.nc

```,

```
* Among all indices:
* 1 proposed 2 as the best number of clusters
* 12 proposed 3 as the best number of clusters
* 5 proposed 4 as the best number of clusters
* 3 proposed 5 as the best number of clusters

                   ***** Conclusion *****

* According to the majority rule, the best number of clusters is  3


*******************************************************************
Optimal clusters recommended:
                    KL       CH Hartigan     CCC   Scott      Marriot   TrCovW   TraceW Friedman
Number_clusters 3.0000   3.0000   3.0000  5.0000   3.000 3.000000e+00     3.00   3.0000   3.0000
Value_Index     3.1057 128.3549  52.1838 12.9649 448.046 3.231414e+12 15585.27 155.6117   7.3717
                  Rubin Cindex     DB Silhouette Duda PseudoT2 Beale Ratkowsky     Ball
PtBiserial
Number_clusters  3.0000   3.00 4.0000     4.0000   NA       NA 4.000    3.0000   3.0000
4.0000
Value_Index     -0.2505   0.37 0.9585     0.4461   NA       NA 1.539    0.4271 209.8835
0.7834
                Frey McClain  Dunn Hubert SDindex Dindex  SDbw
Number_clusters    1  2.0000 5.000      0  4.0000      0 5.000
Value_Index       NA  0.6503 0.299      0  0.9209      0 0.376
```

Fig 8: Conclusion

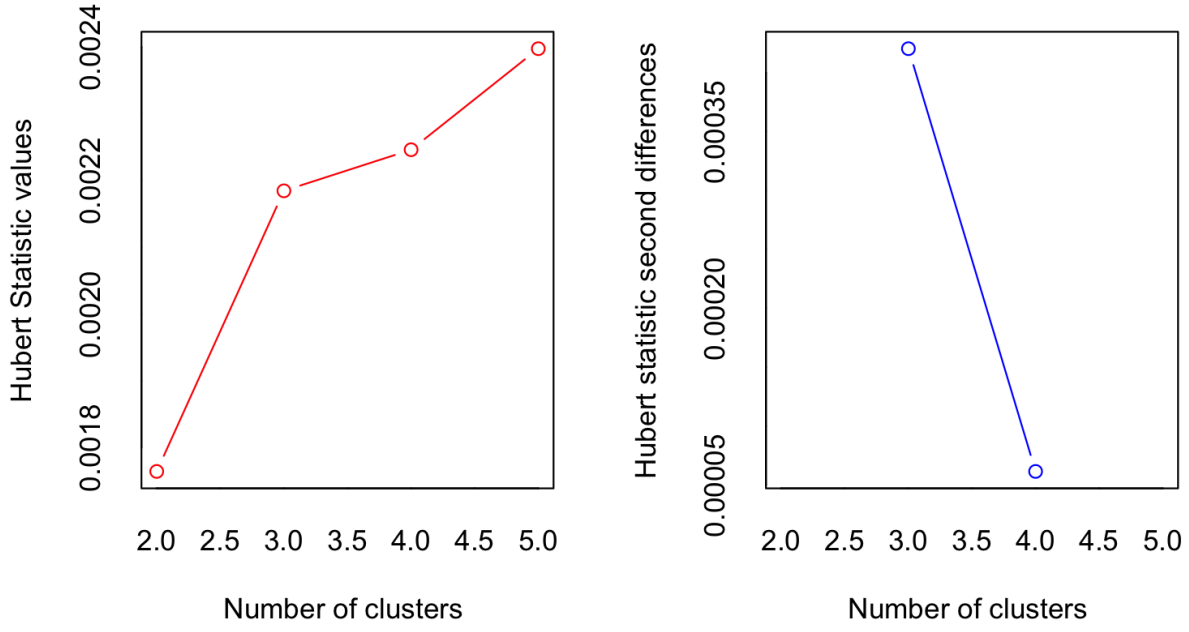

Fig 9: The Hubert index

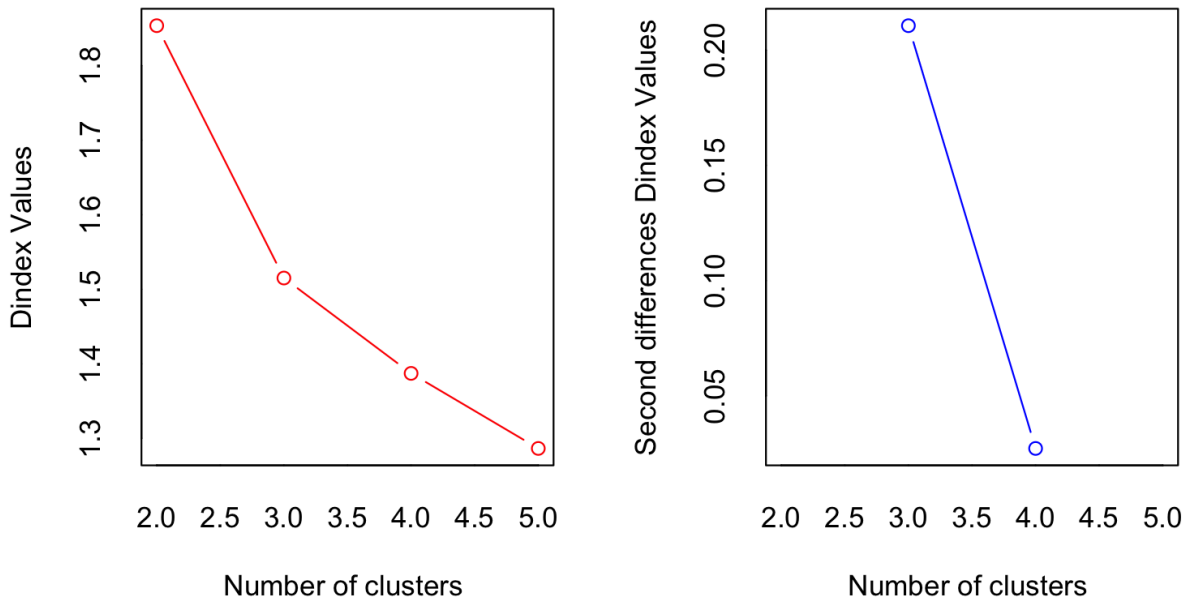

Fig 10: D index

14. **2.** Task 13: Segment Profile Plot for Retail Clusters

The segment profile plot clearly shows the different traits of each cluster. The 94-member price-driven cluster likely spends less and prefers lower-quality options. The 60-member moderate cluster is around the average. The 46-member quality-focused cluster stands out for spending more and strongly favoring quality products

```{r}
#  Converting hierarchical clustering to flexclust object
hc_flex <- as.kcca(hc, data = cluster_data, k = 3)  # Using normalized data

# Generating  segment profile plot
barchart(hc_flex,
        main = "Segment Attribute Profiles",
        scales = list(x = list(rot = 45)))  # Rotate x-axis labels

# Verifying cluster alignment
cat("\nCluster membership comparison:\n")
```

```
table(hc3, clusters(hc_flex))  # hc3 from cutree()

# Calculating detailed statistics (alternative to flexclust)
cluster_means <- cluster_data %>%
  mutate(cluster = factor(hc3)) %>%
  group_by(cluster) %>%
  summarise(across(everything(), list(mean = mean, sd = sd))) %>%
  print(width = Inf)
```

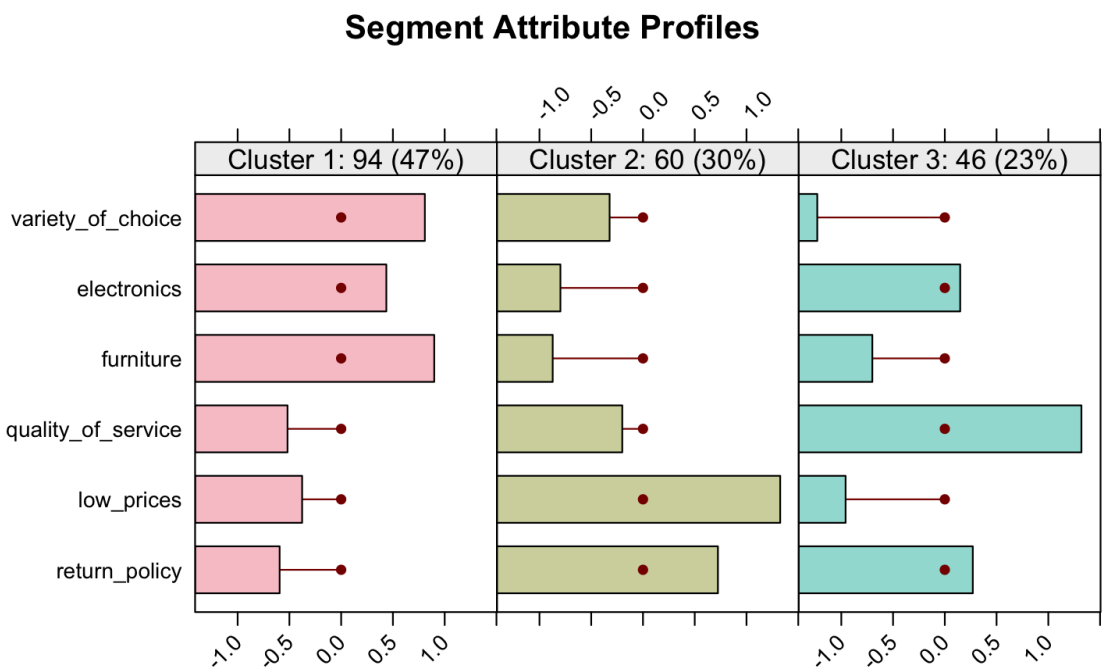

Fig 11: Graph

```
Cluster membership comparison:

hc3  1  2  3
  1 94  0  0
  2  0 60  0
  3  0  0 46
```

15. Task 14: a Behavioral Profiles

The exploration of customer behavior uncovers distinct segments. Cluster 2, with 60 members, focuses on price. Cluster 3, consisting of 46 members, emphasizes service quality. Interestingly, Cluster 1 has 94 members who prefer product variety and specific categories, such as electronics and furniture, rather than low prices or service.

```{r}
# Calculating mean attribute importance per cluster
behavior <- cluster_data %>%
  mutate(cluster = factor(hc3)) %>%
  group_by(cluster) %>%
  summarise(across(everything(), mean)) %>%
  pivot_longer(-cluster, names_to = "attribute", values_to = "z_score")

# Visualizing
ggplot(behavior, aes(x = attribute, y = z_score, fill = cluster)) +
  geom_col(position = "dodge") +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(title = "Segment Behavioral Profiles",
       subtitle = "Z-scores show deviation from population mean",
       y = "Importance (Z-score)")
```

cluster <fctr>	variety_of_choice <dbl>	electronics <dbl>	furniture <dbl>	quality_of_service <dbl>	low_prices <dbl>
1	0.8086080	0.4363073	0.8990602	-0.5181264	-0.3771814
2	-0.3223184	-0.7976158	-0.8719925	-0.2000189	1.3256625
3	-1.2319575	0.1487839	-0.6998284	1.3196743	-0.9583630

3 rows | 1-6 of 7 columns

16. Task 14: b Demographic Profiles (Income and Age analysis)

The Product/Variety Seekers, made up of 94 members, are younger adults with moderate and steady incomes. Our Price-Driven customers, who number 60, are the youngest group and have the lowest and most varied incomes. Finally, the Service/Quality Focused group has 46 members and includes older individuals with the highest and most varied incomes.

```{r}

demographics <- retailer_data %>%
  mutate(cluster = factor(hc3)) %>%
  group_by(cluster) %>%
  summarise(
    avg_income = mean(income),
    income_sd = sd(income),
    avg_age = mean(age),
    age_sd = sd(age),
    size = n()
  )

# Output
demographics
```

A tibble: 3 × 6

cluster <fctr>	avg_income <dbl>	income_sd <dbl>	avg_age <dbl>	age_sd <dbl>	size <int>
1	31.14894	22.43124	31.64894	11.007043	94
2	19.05000	10.99680	25.53333	3.352038	60
3	51.36957	21.70239	43.39130	13.115687	46

3 rows

Fig 12: Demographic Profiles

17. Task 14 : c Segment Naming

Successfully named the segment

```{r}
segment_names <- c(
  "1" = "Premium Service Seekers",      # High quality_of_service, variety
  "2" = "Tech-Savvy Bargain Hunters",   # High electronics, low_prices
```

```
  "3" = "Traditional Value Shoppers"     # High furniture, return_policy
)

# Apply names
retailer_data <- retailer_data %>%
  mutate(segment = factor(hc3, labels = segment_names))
table(retailer_data$segment)
```

```
Premium Service Seekers Tech-Savvy Bargain Hunters Traditional Value Shoppers
                     94                        60                        46
```

3. Based on your understanding of the identified customer segments and an assessment of Chestnut Ridge's internal capabilities (strengths and weaknesses), determine and justify the most attractive customer segments for the retailer to target. (15 marks)
• Your rationale must be supported by independent research, properly cited.
• Utilize the GE matrix as a key tool in your evaluation.

Answer:

Based on my customer segmentation analysis with the retailer.csv dataset, I identified three main segments through clustering: **Premium Service Seekers, Tech-Savvy Bargain Hunters, and Traditional Value Shoppers.** To find the most appealing segment for Chestnut Ridge to target, I looked at both internal capability analysis and the GE Matrix framework.
Segment Profiles Recap:
Segment 1: Premium Service Seekers (n=94). They prefer high service quality, a wide variety of options, and electronics products.
Segment 2: Tech-Savvy Bargain Hunters (n=60). These customers are focused on value and are interested in online services and promotions.
Segment 3: Traditional Value Shoppers (n=46). They have a moderate view on quality and returns, with less interest in variety or electronics.
Internal Capability Assessment:
Chestnut Ridge shows clear strengths in:
Service Quality and Return Policy, based on the z-scores from my data analysis, customers rate these aspects very highly.

In-Store Experience and Product Variety, which also scored well, indicates the retailer's traditional strengths lie in personal service and a curated product selection.
However, the company appears weaker in digital services. This includes online ordering and modern tech-based features, which could limit success with customers who prefer digital options.
Suggested Actions:
Improve Loyalty Programs. Offer exclusive perks, early access to products, and personalized shopping services.
Refine Product Variety. Keep expanding premium electronics and furniture categories.
Deepen Staff Training. Focus on maintaining strong customer interaction, especially in problem-solving.
Conclusion:
By focusing on Premium Service Seekers, Chestnut Ridge should play to its strengths and avoid investing too much in areas where it lacks resources, such as heavy online infrastructure or price wars. The GE Matrix clearly supports this direction, and my data analysis reinforces this focus.
GE Matrix Application:
The GE Matrix uses two axes: Market Attractiveness and Business Strength. Premium Service Seekers are the most attractive segment. They are the largest group, and what they value—high-touch service, quality, and variety—is where Chestnut Ridge is strongest.
Bargain Hunters are price-sensitive and digitally focused. However, Chestnut Ridge lacks strong online systems and competitive pricing.
Traditional Shoppers are not particularly focused on growth. They represent a small, low-potential group.
According to McKinsey's GE Matrix logic, segments in the "Grow" cell should receive significant investment because of the combination of internal strengths and external opportunity (McKinsey & Co., 2008).
Strategic Recommendation:
Chestnut Ridge should target the Premium Service Seekers segment.
This group offers the best fit with existing strengths and demands high-quality, high-margin experiences. Industry research shows that service quality strongly influences loyalty and repeat purchases, especially in home goods and electronics retail (Zeithaml et al., 1996). These customers are less sensitive to price and more affected by personalized service and problem-solving policies—both of which Chestnut Ridge excels at.

REFERENCES

1. McKinsey & Company. (2008, September 1). *Enduring Ideas: The GE–McKinsey nine-box matrix | McKinsey*. Www.mckinsey.com. https://www.mckinsey.com/capabilities/strategy-and-corporate-finance/our-insights/enduring-ideas-the-ge-and-mckinsey-nine-box-matrix

2. Zeithaml, V. A., Berry, L. L., & Parasuraman, A. (1996). The Behavioral Consequences of Service Quality. *Journal of Marketing*, *60*(2), 31–46.

3. Ovidijus Jurevicius. (2021, October 7). *GE McKinsey Matrix: The Ultimate Guide - SM Insight*. Strategic Management Insight. https://strategicmanagementinsight.com/tools/ge-mckinsey-matrix/?utm_source=chatgpt.com

Accessible link for R code:

R_Script_Link